ZAYED MAKANDAR

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SUMMARY

AI/ML Engineer with 1+ year of experience building deep learning and computer vision systems: vision-language document understanding, real-time facial and vital-sign analysis, and RAG voice assistants. Focused on optimising models for real hardware (vLLM on GPU; INT8 ONNX, OpenVINO, llama.cpp on CPU) and deploying them as Dockerised APIs.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Git, Linux
Machine Learning: Regression, Classification, Decision Trees, Random Forest, SVM, k-NN, Naive Bayes, Bagging & Boosting, K-Means, DBSCAN, PCA, Feature Engineering, Model Evaluation
Deep Learning: Backpropagation, CNN, RNN/LSTM/GRU, Autoencoders, GANs, Transformers & Attention, PyTorch, scikit-learn, NumPy, Pandas
Computer Vision: OpenCV, MediaPipe, InsightFace, YOLO (Ultralytics), VLM-based OCR, rPPG
NLP & LLMs: Tokenization, Lemmatization, Word Embeddings, NLTK, spaCy, Hugging Face Transformers, RAG, LangGraph, RAGAS, Whisper
Model Optimisation: ONNX Runtime, TensorRT, INT8/INT4 Quantisation, vLLM, OpenVINO, CUDA
Deployment & MLOps: FastAPI, Docker, PostgreSQL, ChromaDB, MLflow, GitHub Actions

WORK EXPERIENCE

Abhyudyaya Techno Solutions Jun 2025 – Present
AI/ML Engineer *Karnataka, IND*

- Built a CPU-only document-vision pipeline with self-hosted vision-language models (llama.cpp GGUF, OpenVINO INT4) extracting line items from scanned vendor quotations; image-resolution capping cut visual tokens by 67% and latency 3.47×.
- Added validation that recovers truncated model output and rejects extractions whose arithmetic doesn't reconcile, before data reaches purchase orders.
- Built a multi-tenant inference platform serving self-hosted speech-to-text, LLM chat and TTS to AR/VR institutes, with GPU-aware queuing and GPU-to-CPU fallback.
- Designed and taught an AI/ML curriculum covering classical ML, deep learning (CNN, RNN/LSTM, GANs, Transformers), NLP and computer vision, with hands-on lab kits, for AI/ML trainers and students.

PROJECTS

DocIntel: Document Intelligence API | *FastAPI, vLLM, Transformers, Docker* GitHub

- Built a GPU microservice converting PDFs, images and Office files into Markdown/JSON in 90+ languages with a 5B vision-language OCR model and a switchable Transformers/vLLM backend.
- Benchmarked and tuned serving (batching, prefix caching, shared-GPU VRAM budgeting), cutting a 10-page document from 95 s to 13.3 s (7× faster).

Edge-AI Health Monitor | *PyTorch, ONNX Runtime, OpenCV, MediaPipe* GitHub

- Designed a multi-pathway PyTorch model (1D-CNN, 2D-CNN on wavelet scalograms, LSTM, self-attention, fusion layer) feeding 5 risk modules; quantised all 10 models to INT8 ONNX for offline CPU use.
- Built a real-time webcam vision module (MediaPipe FaceMesh, InsightFace, OpenCV) estimating contactless heart rate via rPPG plus blink rate, stress and pallor, fused with IoT vital-sign sensors.

VoiceDesk: AI Front-Desk Assistant | *faster-whisper, ChromaDB, Ollama, FastAPI* GitHub

- Deployed a self-hosted voice receptionist kiosk: local speech-to-text, RAG over an editable ChromaDB knowledge base and a local LLM, and streamed speech replies.
- Improved recognition in a noisy lobby with a wake-word fuzzy matcher and correction layer built from production logs, plus automatic offline TTS fallback.

EDUCATION

Government Engineering College, Gangavathi 2021 – 2025
B.Tech in Computer Science *Karnataka, IND*